No.

Date:

1 Dik : titik awal P (1,1)
titik akhir Q (10,10)
Clipping Xmin : 1
Ymin : 1
Xmax : 7
Ymax : 7

Dit : Clipping Cohen-Sutherland

Jawab : 1. Hitung region code tiap ujung
=> P(1,1) : $1 \le x \le 7$ dan $1 \le y \le 7$ kode P = 0000
Q (10,10) : x > 7 -> Right : 2 : y > 7 -> Top = 8
kode Q : 8+2 1010

2. Periksa trivial accept / reject
=> (kode P | kode Q) ≠ 0 dan (kode P & kode Q) = 0 -> Partially visible

3. Cari titik potong dengan Y =>
=> P -> Q : slope m : (10-1) / (10-1) = 1 -> Y = X
Jika Y = 7 -> X = 7
titik Potong : (7,7)

4. Ganti Q dengan (7,7), hitung kode baru (7,7) di -> 0000
0000 -> trivial accept
Hasil : garis dari (1,1) sampai (7,7)

Big BOSS

2. Dik : $x1 = 1$ Window : $x1 = 1$ Parametrik : ~~$X = x_1 + U$~~
$y1 = 1$ $x_r = 7$ $= x = x_1 + U . dx, y = y_1 + U . dy$
$x2 = 10$ $yb = 1$ dengan $U \in [0,1]$
$y2 = 10$ $yt = 7$

$dx = x^2 - x^1 = 9$
$dy = y^2 - y^1 = 9$

Dit : Algoritma Clipping Liang - Barsky

Jawab : 1. Susun P dan Q
=> $P_1 = -dx = -9, q1 = x1 - x1 = 1 - 1 = 0$
$P_2 = dx = 9, q2 = xr - x1 = 7 - 1 = 6$
$P3 = -dy = -9, q3 = y1 - yb = 1 - 1 = 0$
$P4 = dy = 9, q4 = yt - y1 = 7 - 1 = 6$
Inisiasi $u1 = 0$ $u2 = 1$

2. Proses tiap batas
=> $-k = 1 : P1 = -9, q1 = 0 \rightarrow r = 0 / (-9) = 0$
$\rightarrow P < 0 \rightarrow U_1 = max(U1, r) = max(0, 0) = 0$

$-k = 2 : P2 = 9, q2 = 6 \rightarrow r = 6/9 = 2/3 \approx 0,666667$
$\rightarrow P > 0 \rightarrow U_2 = min(u2, r) = min(1, 2/3) = 2/3$

$-k = 3 : P3 = -9, q3 = 0 \rightarrow r = 0 \rightarrow P < 0 \rightarrow U1 = max(0,0) = 0$

$-k = 4 : P4 = 9, q4 = 6 \rightarrow r = 6/9 = 2/3 \rightarrow P > 0 \rightarrow U2 = min$
$(2/3, 2/3) = 2/3$

3. Hitung titik akhir baru
=> $(x', y') = (x' + u' . dx, y' + u' . dy) = (1 + 0.9, 1 + 0.9) = (1, 1)$
$(x^2, y^2) = (1 + (2/3) . 9, 1 + ((2/3) . 9) = (1 + 6, 1 + 6) = (7, 7)$
$= (1,1) \rightarrow (7,7)$